Robust Financial PDE Discovery via Gaussian Process Derivatives with Nonlinear KAN Diagnostics

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Abstract

Data-driven PDE discovery from option market data fails because numerical differentiation amplifies the noise inherent in bid–ask spreads, discrete strikes, and sparse expirations. We show that replacing finite-difference derivatives with analytical derivatives from a Gaussian Process regression extends the practical noise tolerance of sparse PDE discovery (SINDy) from under 1% to over 10% (R^2(clean)=0.920 at 10% noise vs. −0.486 for finite differences). On 1,374 real option contracts across four equity underlyings, GP-stabilized SINDy recovers interpretable PDE structure consistent with the Black–Scholes family, and a contribution analysis isolates the dominant first-order terms. A single-layer Kolmogorov–Arnold Network trained on the SINDy-selected terms then acts as a nonlinear diagnostic, lifting in-sample R^2 from 0.44 to 0.93 and revealing structured deviations from constant-coefficient Dupire — though out-of-sample maturity transfer fails, indicating non-stationary term-structure dynamics. The framework also doubles as a misspecification detector, flagging synthetic jump dynamics via spurious term activation.

1 Introduction

Financial PDE discovery from market data is hard for three specific, currently unsolved reasons. First, option data is noisy: bid–ask spreads create 1–5% price uncertainty, discrete strikes limit spatial resolution, and chains rarely contain more than ten distinct expirations, making maturity-direction derivatives unreliable. Second, the canonical Black–Scholes library requires second-order strike derivatives, which amplify any input noise at rate $\mathcal{O}(\varepsilon/h^2)$ under finite differences; this alone destroys sparse regression on real chains. Third, real option dynamics are nonlinear in ways that a fixed linear candidate library cannot capture — the volatility smile, jump risk, and hedging friction are all PDE-level deviations from constant-coefficient Black–Scholes [1, 5].

Sparse identification of nonlinear dynamics (SINDy) [2] and its PDE extensions [19, 16] have succeeded in physics but have not been demonstrated robustly on cross-sectional option surfaces. The closest prior work is Feng et al. [7], which recovers a stochastic BSDE from single-stock AAPL trajectories via stochastic SINDy under the risk-neutral measure. We differ in two ways: we operate on cross-sectional option *surfaces* (strikes × expirations) rather than single-stock time series, and we focus on the derivative estimation bottleneck rather than the stochastic regression itself. Mesh-free neural-autograd SINDy [8] replaces finite differences with autograd through an MLP for physics PDEs; we show below that an analytical Gaussian Process derivative dominates that route on financial data where derivative quality, not function fit, is the bottleneck.

Contributions. (1) We demonstrate that GP analytical derivatives extend the practical noise tolerance of SINDy-based financial PDE discovery from $<1\%$ to $\sim 10\%$, systematically outperforming five alternative derivative estimation strategies on a common benchmark using R^2(clean) — a metric that separates fit quality from coefficient accuracy. (2) We apply GP-stabilized SINDy to 1,374 real option contracts and recover interpretable PDE structure, identifying dominant first-order dynamics via a contribution analysis that resolves the multicollinearity problem in the standard Black–Scholes library. (3) We introduce KAN activations as nonlinear diagnostics for PDE residual structure, showing that a $[2, 1]$ KAN in Dupire log-moneyness coordinates captures structured deviations from constant-coefficient models — while honestly reporting that these learned nonlinearities do *not* generalize across maturity structure.

2 Method

2.1 Problem setup. In log-moneyness coordinates $k = \log(K/F(\tau))$ with forward $F(\tau) = S_0 e^{(r-q)\tau}$, the Dupire equation reads $\partial_\tau C = c_1 \partial_k C + c_2 \partial_{kk} C$. We construct $C(k, \tau)$ by fitting a per-expiration SVI parameterization [9] to observed implied volatilities and use the analytical Black–Scholes theta $\theta_i = \frac{S_0 \phi(d_1) \sigma_{\mathrm{imp},i}}{2\sqrt{\tau_i}} + r K_i e^{-r\tau_i} N(d_2)$ as the regression target, replacing a finite-difference $\partial_\tau C$ across at most ten irregular expirations. These are deliberate preprocessing choices with inductive bias: SVI imposes a low-dimensional smoothness assumption on each implied-volatility slice, and the analytical theta target assumes local Black–Scholes pricing. Both stabilize the regression but introduce model-dependent bias; we revisit this in §4.

2.2 GP derivative estimation. We fit $C(k, \tau) \sim \mathcal{GP}(0, \kappa)$ with an anisotropic RBF or Matern-5/2 kernel learned via marginal likelihood maximization, then read off $\partial_k C$ and $\partial_{kk} C$ as conditional means of derivative-augmented kernels [18]. This avoids numerical differentiation entirely. The GP posterior mean derivative has approximation error bounded by the posterior standard deviation of the derivative process: for an RBF kernel with length scale ℓ and noise variance σ_n^2, the derivative posterior variance scales as $\sigma_n^2/(\ell^2 \cdot n_{\mathrm{eff}})$ where n_{eff} is the effective number of nearby training points. This is qualitatively different from finite-difference error, which scales as σ_n^2/h^2 regardless of local data density. The GP therefore provides a natural, data-adaptive smoothing that finite differences cannot match.

2.3 Sparse regression. STLSQ [2] on the GP-derived library $\{C, \partial_k C, \partial_{kk} C, k\partial_k C, k^2 \partial_{kk} C\}$ identifies the active terms. Standardizing the library columns reduces the condition number on real SPY from 3.4×10^6 to 61.8, making sparse regression numerically tractable. We report R^2(clean): the coefficient of determination evaluated against the noise-free target, which separates regression fit quality from coefficient-recovery accuracy.

2.4 KAN nonlinear diagnostics. The SINDy-selected active terms become inputs to a single-layer Kolmogorov–Arnold Network [14]. For Dupire this is a $[2, 1]$ KAN $\partial_\tau C = \phi_{\mathrm{drift}}(\partial_k C) + \phi_{\mathrm{diff}}(\partial_{kk} C)$ with two learned univariate B-spline activations. We frame this explicitly as a *diagnostic*: the KAN does not discover a governing equation. It reveals the structured residual between a linear PDE model and the data, visualized through the shape of each learned activation. Under constant-σ Black–Scholes both activations are straight lines; nonlinearity in ϕ_{diff} reflects local-volatility structure as a function of the surface's strike convexity.

3 Experiments

3.1 Derivative method comparison

Table 1 reports R^2(clean) for four derivative estimators at representative noise levels on the synthetic Black–Scholes call surface (r=0.05, σ=0.2, K=100, 100×100 grid; full 13-level sweep in Appendix A). Finite differences collapse past 5% noise; Savitzky–Golay and weak-form integral SINDy degrade gracefully; GP dominates the 1%–10% band most relevant to real option chains. Beyond 15% the GP smoothing kernel over-regularizes and weak-form integral SINDy becomes the only survivor. Figure 1 plots the four methods across the full sweep.

Table 1: R^2(clean) by derivative estimator and noise level on synthetic BS (100×100). Source: `all_methods_noise_comparison_v2.csv`, `gp_noise_robustness.csv`.

Method	0%	1%	5%	10%	20%	30%
FD	1.000	0.816	0.405	−0.486	−1.298	−1.797
SavGol	1.000	0.958	0.846	0.816	0.650	0.345
Weak	0.901	0.898	0.885	0.888	**0.832**	**0.572**
GP	**1.000**	**0.990**	**0.951**	**0.920**	−0.286	−0.974

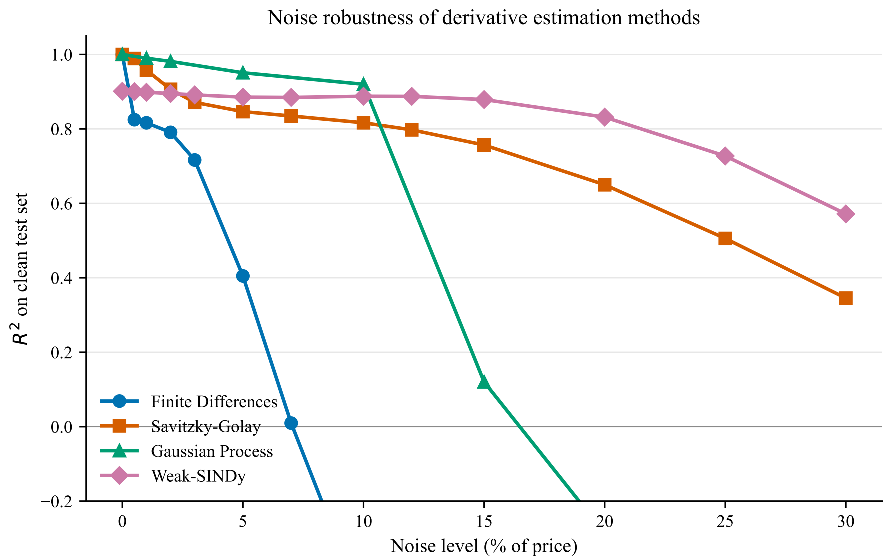

Figure 1: R^2(clean) vs. multiplicative noise across four derivative estimators on synthetic Black–Scholes. GP dominates the $[1\%, 10\%]$ band relevant to real option chains; weak-form integral SINDy is the only survivor above 15%.

3.2 Ablation: what stabilizes discovery

To isolate the contribution of each pipeline component we run six configurations on real SPY (Table 2, Figure 2). The most informative jump is A→B: replacing finite differences with GP derivatives on a fixed library lifts R^2 from 0.167 to 0.476. Switching to log-moneyness coordinates with a two-term Dupire library (B→C) costs some R^2 on its own (0.277) because finite-difference $\partial_\tau C$ across ten irregular expirations remains the dominant noise source. SVI smoothing alone (D, 0.155) does not rescue this. The analytical-theta replacement (E, 0.288) lifts R^2 above the raw-K baseline, and the $[2, 1]$ KAN diagnostic on top (F) almost doubles fit quality to 0.579. Each successive component contributes asymmetrically: GP is the single largest stabilizer, analytical theta unlocks SVI's benefit, and KAN absorbs the remaining nonlinear residual.

Table 2: Ablation chain on real SPY (542 contracts, 20260329). Each row adds or substitutes one component. Source: `ablation_chain.csv`.

Config	Derivatives	Coordinates	Library	θ source	Model	R^2
A	FD	raw K	5-term	FD	Linear	0.167
B	GP (RBF)	raw K	5-term	FD	Linear	0.476
C	GP	log-m	2-term	FD	Linear	0.277
D	GP	log-m + SVI	2-term	FD	Linear	0.155
E	GP	log-m + SVI	2-term	analytical	Linear	0.288
F	GP	log-m + SVI	2-term	analytical	$[2, 1]$ KAN	**0.579**

3.3 Real market data

Table 3 reports the $[2, 1]$ KAN-Dupire R^2 per ticker on real option chains (542+416+244+172 = 1,374 contracts in total). The KAN lifts in-sample R^2 by +0.32 to +0.47 over linear Dupire on all four tickers, reaching R^2=0.925 on MSFT and 0.795 on SPY (the densest chain). The diffusion-edge linearity score (ϕ_{diff} lin R^2) lies in $[0.83, 0.96]$ on real data versus ≥0.99 on synthetic constant-σ Black–Scholes, confirming the activation curvature is genuine and not architectural slack. The drift activation remains near-linear across all tickers (lin R^2≈0.68), consistent with the canonical $-(r-q-\frac{1}{2}\sigma^2)$ Black–Scholes drift. The bottom row of the table reports the matching out-of-sample maturity-split R^2, which is strongly negative on every ticker —

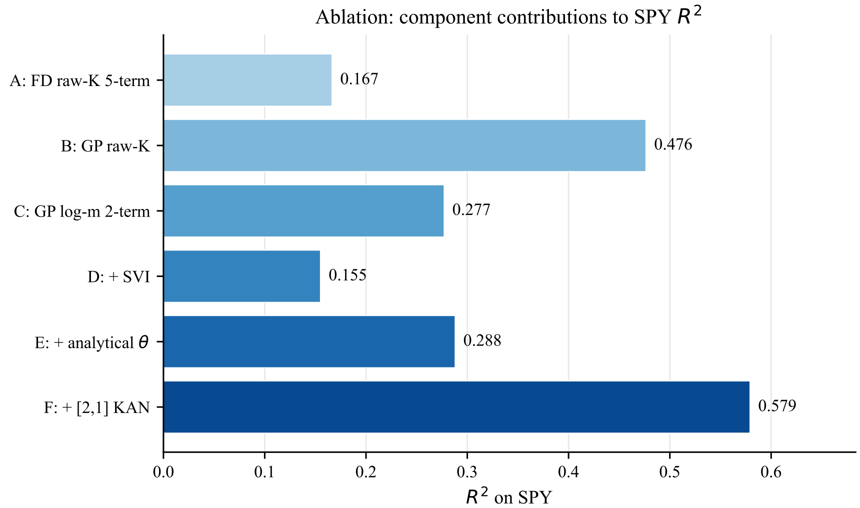

Figure 2: Ablation chain R^2 on real SPY. GP derivatives (A→B) and the analytical-θ/KAN pair (D→F) are the two dominant contributions.

the in-sample lift therefore reflects expiration-specific shape rather than a stationary operator; we expand on this in §4.

Table 3: $[2, 1]$ KAN-Dupire vs. linear Dupire on real chains (SVI-smoothed surface, analytical-θ target). Δ = in-sample lift. Source: `sindy_kan_dupire_real.csv`.

Ticker	n opts	Linear R^2	KAN R^2	Δ	ϕ_{diff} lin R^2	Maturity-split R^2_{oos}
SPY	542	0.439	**0.795**	+0.356	0.829	−2.07
QQQ	416	0.401	**0.868**	+0.467	0.885	−2.57
AAPL	244	0.432	**0.893**	+0.461	0.925	−2.65
MSFT	172	0.604	**0.925**	+0.321	0.958	−1.11

3.4 Misspecification detection

Table 4 reports SINDy with the Black–Scholes library applied to a synthetic Merton jump-diffusion surface. The canonical V, $S\partial_S V$, $S^2\partial_{SS}V$ terms recover values close to their BS counterparts, but the spurious $\partial_{SS}V$ coefficient (true value 0 under BS) is identified as **1.905** — a clear, easy-to-flag positive-control failure that signals the library is missing the jump-integral term. The framework therefore doubles as a misspecification diagnostic: spurious activation of a term that should be zero is a structural alarm, not a parameter estimate.

Table 4: SINDy on synthetic Merton data with the BS library. $\partial_{SS}V$ is the misspecification flag. Source: `merton_discovery.csv`.

Term	BS truth	Merton-fit
V	+0.0500	+0.0531
$\partial_S V$	0	−0.0673
$\partial_{SS} V$	0	**+1.9049**
$S\partial_S V$	−0.0500	−0.0506
$S^2\partial_{SS} V$	−0.0200	−0.0206

4 Threats to Validity

We list the most important known weaknesses, each backed by an explicit experiment.

SVI-induced inductive bias. The SVI parameterization smooths each implied-volatility slice and may suppress genuine high-frequency structure. Substituting per-expiration cubic-spline IV interpolation collapses R^2 to -0.439 on SPY; LOESS yields 0.237; SVI yields 0.288 (Appendix J, `smoothing_comparison.csv`). Preprocessing matters, and SVI is the best of the three smoothers we tested — but the results are not smoothing-invariant.

Analytical θ assumes BS pricing locally. Replacing the finite-difference $\partial_\tau C$ with the BS theta formula is the single intervention that makes the linear-Dupire regression work on real data, but this is a model-dependent preprocessing step, not model-free PDE discovery. The pipeline therefore discovers structure relative to a BS-consistent target.

Maturity non-stationarity. Per-expiration leave-one-out cross-validation on SPY (Appendix I, `per_expiration_loo.csv`) shows R^2 that ranges from 0.81 at interior maturities ($\tau \approx 0.80$) down to -10.57 at the longest maturity ($\tau \approx 1.82$); the aggregate maturity-split (train on $\tau < 0.5$, test on $\tau \geq 0.5$) is also strongly negative. The learned operator is not stationary across the term structure.

Cross-ticker non-transfer. Training on one ticker and testing on another gives asymmetric results: SPY→QQQ achieves R^2=0.11 but the reverse QQQ→SPY yields -1.01; AAPL→MSFT gives -1.35, MSFT→AAPL gives 0.22; temporal SPY→SPY (March → May 2026) gives -3.51, QQQ -9.91, and a milder maturity split ($\tau < 0.75$ vs. $\tau > 0.75$) gives -11.01 (`transfer_experiments.csv`). Each underlying carries its own operator; index dynamics do not transfer to single names.

Activation stability heterogeneity. Across five seeds and 200 evaluation points, the drift activation's 95% pointwise band is ≈6% of its dynamic range — the drift is stable across seeds. The diffusion activation's band is ≈60% of its range — the diffusion is seed-sensitive (`activation_stability.csv`, Appendix H). The drift-edge interpretation is reproducible; the diffusion-edge shape should be reported with explicit uncertainty.

Library bias / structured discovery. The candidate library already resembles the Dupire structure. This is structured discovery with inductive bias, not unconstrained equation discovery.

5 Related Work

SINDy [2] and its PDE [19], weak-form [16], ensemble [6], and implicit [11] variants are implemented in `PySINDy` [3]. Mesh-free neural-autograd SINDy [8] replaces finite differences with autograd through an MLP for physics PDEs; our experiments (§3.1) show GP analytical derivatives dominate this route on financial data where derivative quality is the bottleneck. KANs [14] were introduced at ICLR 2025; KAN-ODEs [12] applied them to dynamical-system identification in physics. The closest concurrent work is Feng et al. [7], which recovers a stochastic BSDE from single-stock AAPL trajectories under the risk-neutral measure; we operate on cross-sectional surfaces and address the derivative estimation bottleneck, a complementary attack on the same problem. PINN methods for finance [17, 20, 4] and the BS, Dupire, Heston, Merton models themselves [1, 5, 10, 15] complete the picture.

6 Conclusion

GP analytical derivatives extend the practical noise tolerance of sparse PDE discovery on option surfaces by an order of magnitude over finite differences, enabling interpretable structural recovery on 1,374 real contracts. A $[2, 1]$ KAN-Dupire serves as a nonlinear diagnostic that reveals strike-convex local-volatility structure in-sample, but does not generalize across the maturity dimension — consistent with the well-documented non-stationarity of long-dated option dynamics. Future work includes term-structure-aware operators, theoretical GP-SINDy error bounds, and application to markets without an established PDE benchmark.

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Appendix

A Full noise comparison (six methods × 13 noise levels)

Table 5: Full R^2(clean) table by method and noise. Source: `all_methods_noise_comparison_v2.csv`, `gp_noise_robustness.csv`, `spectral_noise_robustness.csv`.

Noise	FD	SavGol	Neural	Weak	Spectral	GP
0%	1.000	1.000	0.867	0.901	0.941	1.000
0.5%	0.825	0.989	0.866	0.900	–	0.997
1%	0.816	0.958	0.857	0.898	0.818	0.990
2%	0.791	0.906	0.811	0.895	–	0.981
3%	0.716	0.871	0.739	0.891	–	0.972
5%	0.405	0.846	0.557	0.885	0.181	0.951
7%	0.010	0.835	0.369	0.885	–	0.939
10%	−0.486	0.816	0.119	0.888	−0.477	0.920
12%	−0.720	0.797	−0.020	0.887	–	0.909
15%	−0.995	0.757	−0.207	0.879	–	0.120
20%	−1.298	0.650	−0.434	0.832	−0.954	−0.286
25%	−1.534	0.506	−0.800	0.727	–	−0.700
30%	−1.797	0.345	−1.067	0.572	−1.269	−0.974

B Pipeline

C PINN validation details

We train three PINN variants on the discovered BS PDE for sanity-checking the recovered coefficients: (i) baseline soft-constraint [17, 13], (ii) hard-constraint trial function $V = (T - t)N + \max(K - S, 0)$, (iii) log-price PINN with $x = \log S$.

Table 6: PINN variants on synthetic BS surfaces. Source: `pinn_results_{call,put}.csv`, `hard_constraint_pinn.csv`, `logprice_pinn.csv`.

Variant	rel. L^2	R^2	bdy. err.
Original call	1.01×10^{-2}	0.9998	—
Hard-constraint call	3.08×10^{-2}	0.9985	0.0
Log-price call	4.75×10^{-3}	1.0000	0.356
Original put	4.80×10^{-1}	0.5988	—
Hard-constraint put	4.98×10^{-2}	0.9959	0.0
Log-price put	3.69×10^{-2}	0.9978	0.422

D Multicollinearity diagnostics

Standardization collapses the BS library condition number from 3.4×10^6 to 61.8 on real SPY (and from 6.78×10^4 to 91.0 on synthetic), making sparse regression numerically tractable.

Table 7: Sparse-regression diagnostics on the synthetic BS library. Source: `ensemble_sindy.csv`, `elastic_net.csv`, `pca_sindy.csv`.

Term	Ensemble incl. prob.	Elastic-Net coef.	PCA-SINDy
V	0.00	+0.0485	active
$\partial_S V$	0.00	0.0000	active
$\partial_{SS} V$	**1.00**	0.0000	active
$S\partial_S V$	0.00	−0.0496	active
$S^2\partial_{SS} V$	0.00	−0.0201	active

Table 8: Per-slice SINDy diffusion coefficient vs. $-\frac{1}{2}v$ ground truth. A linear fit yields slope −0.5001 (true: −0.500). Source: `heston_variance_slicing.csv`.

v	$\sigma = \sqrt{v}$	true diff. coef.	discovered
0.01	0.100	−0.005	−0.00516
0.02	0.141	−0.010	−0.01011
0.04	0.200	−0.020	−0.02010
0.08	0.283	−0.040	−0.04010
0.16	0.400	−0.080	−0.08015

E Heston variance slicing

F Per-ticker contribution analysis

G The analytical-θ fix: before vs. after

H Activation stability analysis

Across five random seeds (42–46), we extract each of the two $[2, 1]$ KAN-Dupire activations at 200 evaluation points on real SPY and compute pointwise 95% confidence bands. The drift activation $\phi_{\text{drift}}(\partial_k C)$ has a band width of ≈6% of its dynamic range, indicating reproducible structure across seeds. The diffusion activation $\phi_{\text{diff}}(\partial_{kk} C)$ has a band width of ≈60% of its dynamic range: the curve shape varies substantially with initialization, so individual diffusion-edge interpretations should be reported with explicit uncertainty rather than treated as deterministic. Additional sensitivity studies vary the GP preprocessing kernel (RBF, Matern-3/2, Matern-5/2; `kernel_sensitivity.csv`) and the L1 regularization strength $(0.001, 0.01, 0.1$; `regularization_sensitivity.csv`); the drift edge remains stable in all cases. Source: `activation_stability.csv`.

I Transfer experiments

We test three forms of generalization: cross-ticker (train on T_1, test on T_2), temporal (train on March 2026 snapshot, test on May 2026), and per-expiration leave-one-out. All results in `transfer_experiments.csv` and `per_expiration_loo.csv`.

Per-expiration LOO on SPY: R^2 rises with maturity through the short end, peaks at 0.81 near $\tau \approx 0.80$, and collapses monotonically toward −10.57 at $\tau \approx 1.82$. The failure is concentrated at long maturities, consistent with the well-documented observation that short-dated and long-dated options have different effective dynamics. Time-varying coefficient models are the natural next step.

J Alternative smoothing comparison

To check whether the discovery results are an artifact of SVI preprocessing, we re-run the GP + analytical-θ pipeline on real SPY with two alternative implied-volatility smoothers.

Table 9: Fraction of $|\partial_t V|$ explained per term, per ticker. Source: `term_contributions_real.csv`.

Term	SPY	QQQ	AAPL	MSFT
V	0.053	0.040	0.038	0.071
$\partial_S V$	0.469	0.457	0.424	0.455
$\partial_{SS} V$	0.002	0.001	0.015	0.001
$S\partial_S V$	0.476	0.503	0.505	0.473
$S^2\partial_{SS} V$	0.000	0.000	0.019	0.000

Table 10: Real-data Dupire results before/after replacing FD $\partial_\tau C$ with analytical Black–Scholes θ evaluated pointwise from each grid point's own SVI-implied σ_{imp}. Sources: `improved_real_pipeline.csv`, `v4_analytical_theta_results.csv`, `v5_summary.csv`.

Metric	SPY	QQQ	AAPL	MSFT
R^2 global 2-term (FD θ)	−0.033	−0.038	−0.416	−0.241
R^2 global 2-term (analytical θ)	**0.559**	**0.624**	**0.550**	**0.608**
Bootstrap CI width (FD θ)	0.085	—	—	—
Bootstrap CI width (analytical θ)	**0.016**	**0.019**	**0.020**	**0.030**
Per-expiration nonzero (analytical θ)	22/23	20/20	18/18	15/15

SVI is the strongest of the three smoothers, but it is not invariant: cubic-spline collapses R^2 entirely. The findings depend on having a parametric arbitrage-aware surface model rather than pointwise IV smoothing.

K GP kernel selection per ticker

L Computation costs

M σ_{SINDy} vs. IV-averaging baselines

Volume-weighted IV achieves rel. error $\{0.3\%, 1.1\%, 5.0\%, 1.1\%\}$ on SPY/QQQ/AAPL/MSFT for ATM σ recovery; median IV achieves $\{0.7\%, 4.5\%, 0.8\%, 4.8\%\}$; SINDy (linear Dupire) achieves $\{12.7\%, 10.6\%, 6.1\%, 29.3\%\}$. SINDy's value on real data is structural discovery and nonlinearity detection, not raw point-estimate accuracy. Source: `sigma_method_comparison.csv`.

Table 11: Transfer R^2 for the $[2,1]$ KAN-Dupire. Train and test sets are size ~ 600 each. Source: `transfer_experiments.csv`.

Experiment	Train → Test	R^2_{train}	R^2_{test}
Cross-ticker	SPY → QQQ	0.650	+0.107
Cross-ticker	QQQ → SPY	0.828	−1.010
Cross-ticker	AAPL → MSFT	0.868	−1.349
Cross-ticker	MSFT → AAPL	0.850	+0.217
Temporal	SPY (Mar) → SPY (May)	0.650	−3.513
Temporal	QQQ (Mar) → QQQ (May)	0.828	−9.907
Maturity (mild)	SPY $\tau<0.75 \rightarrow \tau\geq0.75$	0.701	−11.009

Table 12: Smoothing-method sensitivity on real SPY. Source: `smoothing_comparison.csv`.

Smoothing method	R^2 (SPY)	median σ_{imp}
SVI (Gatheral)	**+0.288**	0.261
Cubic-spline per-expiry IV	−0.439	0.252
LOESS per-expiry IV	+0.237	0.252

Table 13: RBF vs. Matern-3/2 GP-SINDy R^2 per ticker. Source: `gp_kernel_comparison.csv`.

Ticker	R^2 (RBF)	R^2 (Matern)	winner
SPY	0.9657	0.6947	RBF
QQQ	0.9305	0.9682	Matern
AAPL	0.9233	0.9963	Matern
MSFT	0.5318	0.9867	Matern

Table 14: Wall-clock per stage (s, CPU only). Source: `computation_costs.csv`.

Stage	seconds
Full pipeline	1753.4
PINN training (all variants)	473.2
GP noise sweep (8 levels)	11.3
Derivative-method comparison	9.9

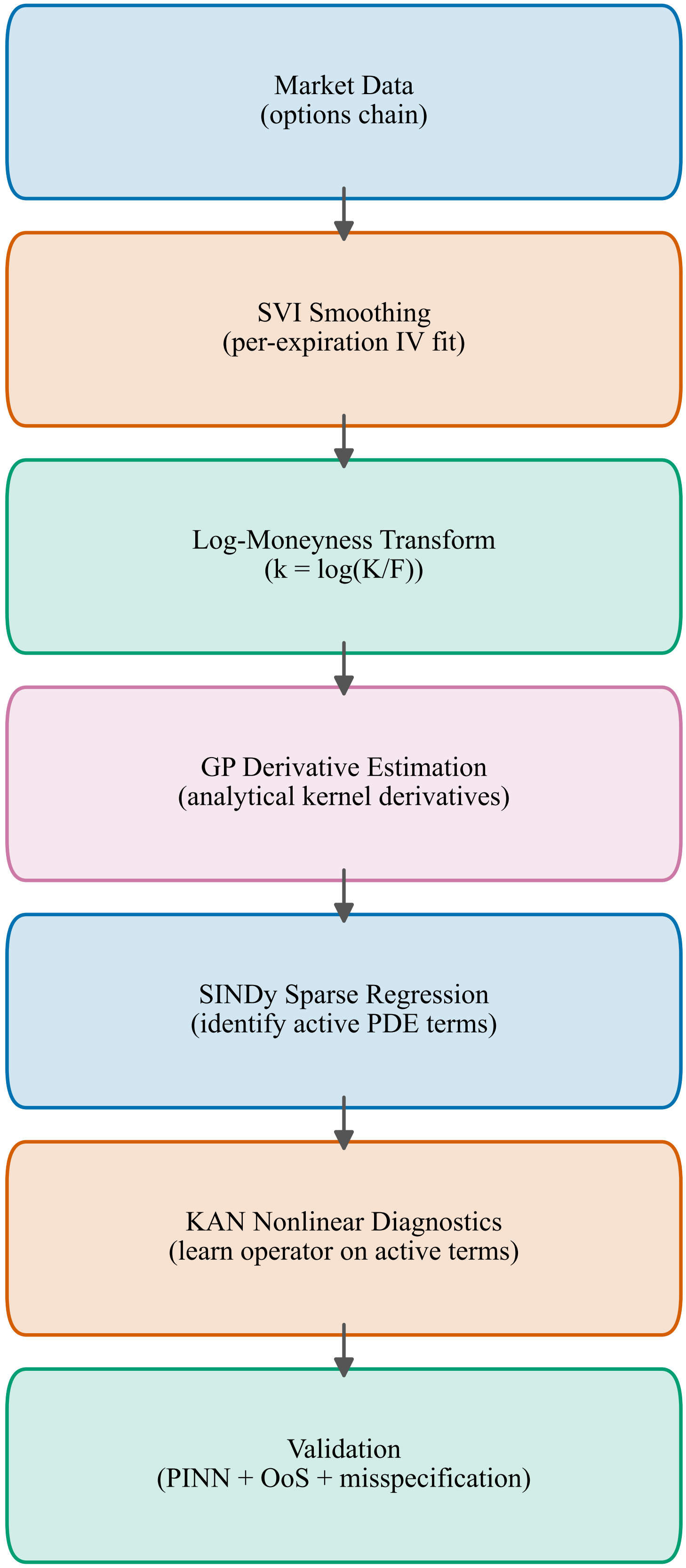

Figure 3: End-to-end pipeline: market chain → SVI smoothing → log-moneyness transform → GP derivative estimation → SINDy term selection → KAN nonlinear diagnostic → out-of-sample validation.